# 📝 Unit 8 Guided Notes — Transformations

Name _____ Date _____ Period _____

Slide it. Flip it. Turn it. Resize it. Then describe what changed and what stayed the same.

**🎯 By the end of this unit, I can…**

- Perform translations, reflections, rotations, and dilations on polygons
- Describe a sequence of transformations between two figures
- Tell congruence from similarity based on the transformations used

**Words to know**

- **Transformation** — a move or change of a shape.
- **Pre-image** — the shape before the move.
- **Image** — the shape after the move. Its points get a prime mark: A′.
- **Rigid transformation** — a move that keeps the same size and shape.
- **Congruent** — same size and same shape.
- **Similar** — same shape. The size can be different.
- **Scale factor (k)** — the number you multiply by in a dilation.

*Standards: NC.8.G.2 · NC.8.G.3 · NC.8.G.4*

Fill in the blanks as we go. For every example and problem, **write each step on its own line** in the space given, and box your final answer.

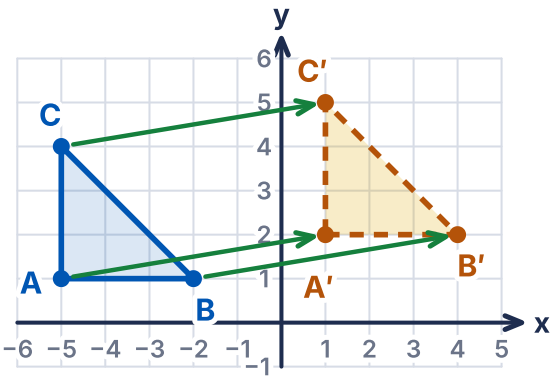
## 1 Transformation Vocabulary

Before we slide, flip, turn, or resize anything, we need shared language. Every transformation moves points by a rule — and we always describe what changed and what stayed the same.

📌 **Transformation:** A rule that moves, flips, turns, or resizes every point of a figure in a predictable way.

📌 **Pre-image:** The *original* figure, BEFORE the transformation. Vertices use plain letters: A, B, C.

📌 **Image:** The *new* figure, AFTER the transformation. Vertices use a prime mark: A′, B′, C′ (read "A prime").

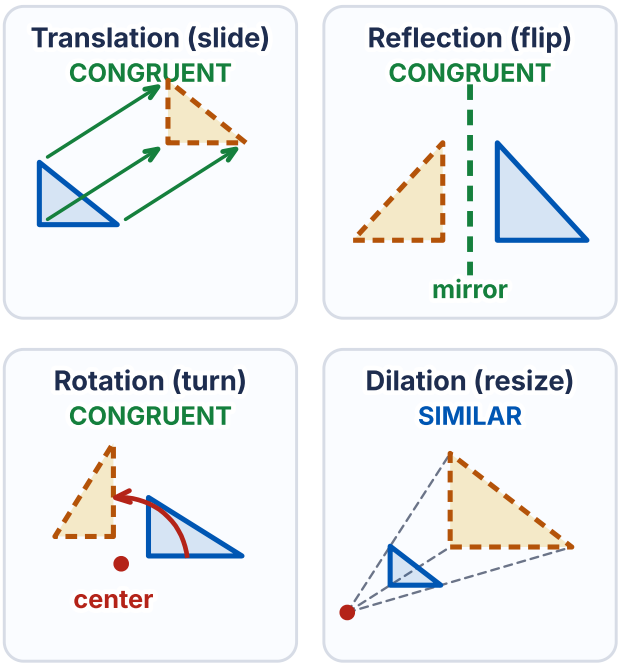



**Pre-image ABC** → **image A′B′C′**. Each point A goes to its partner A′.

## Two Families of Transformations

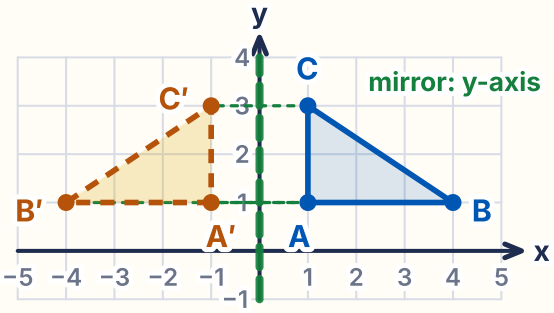
| Family | What Stays the Same | Types | Result |
|---|---|---|---|
| **Rigid** (size stays the same) | Size AND shape | Translation, Reflection, Rotation | Pre-image & image are **CONGRUENT** |
| **Non-Rigid** | Shape only (size changes) | Dilation | Pre-image & image are **SIMILAR** |

💡 **Big idea:** If size stays the same, it's **congruent**. If size changes but shape stays the same, it's **similar**.




## Quick Check

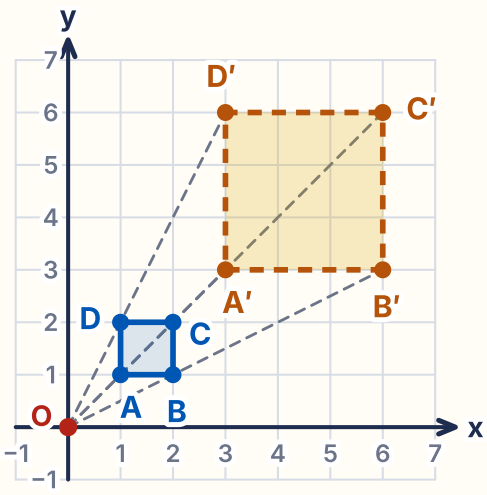
**Q1** A triangle is reflected over the y-axis. The pre-image and image are...




**blue** = pre-image · **gold dashed** = image

- ○ Similar but not congruent
- ○ Congruent
- ○ Neither

**Q2** **A square is dilated by a scale factor of 3. The pre-image and image are...**




**Rule: (x, y) → (3x, 3y)**

**blue** = pre-image · **gold dashed** = image

- ○ Similar but not congruent
- ○ Congruent
- ○ Neither

## 2 Translations (Slides)

📌 **Translation:** Every point slides the *same distance* in the *same direction*. The shape is NOT rotated or flipped — it just moves.

### The Rule

$$(x,\ y) \rightarrow (x + h,\ y + k)$$

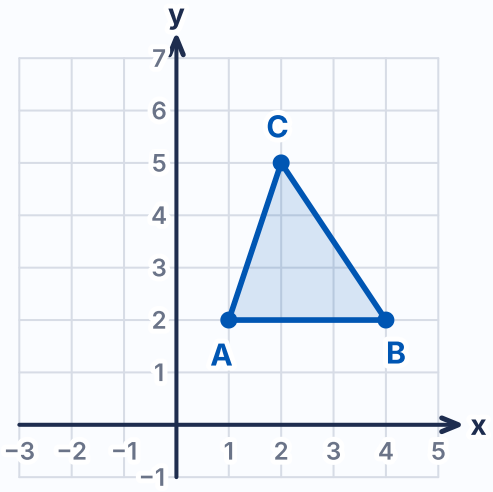
**h** = horizontal shift | **+** moves right, **−** moves left
**k** = vertical shift | **+** moves up, **−** moves down

💡 **Tip:** Add the numbers. + moves right or up. − moves left or down.

## Worked Example

**Example 1: Triangle ABC has vertices A(1, 2), B(4, 2), C(2, 5). Translate by (x − 3, y + 1).**

*Pre-image △ABC (blue).*




**Rule: (x, y) → (x − 3, y + 1)**

**blue** = pre-image · **gold dashed** = image

**Step 1 — Identify h and k:** _____

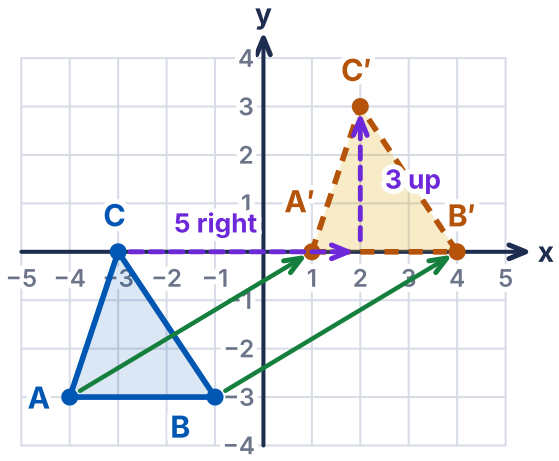
**Step 2 — Apply rule to A(1, 2):** _____

**Step 3 — Apply rule to B(4, 2):** _____

**Step 4 — Apply rule to C(2, 5):** _____

✓ **Image vertices:** _____

## Visual




**Rule: (x, y) → (x + 5, y + 3)**

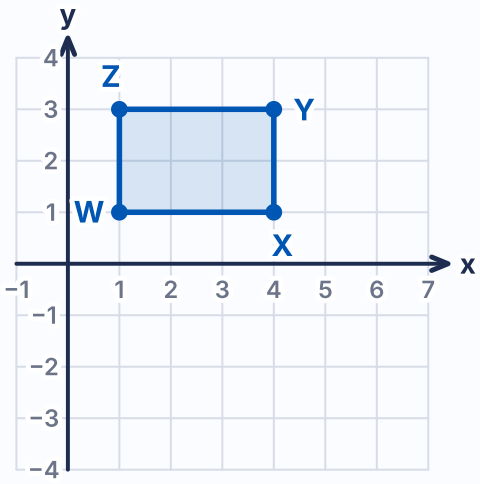
**blue** = pre-image · **gold dashed** = image

Rule (x + 5, y + 3): **h = +5** moves every point 5 right, **k = +3** moves it 3 up. Every arrow is the same slide.

## Practice

**P1** **Rectangle WXYZ has vertices W(1, 1), X(4, 1), Y(4, 3), Z(1, 3). Translate the rectangle by the rule (x + 2, y − 4). What are the coordinates of W′?**

- ○ (3, −3)
- ○ (−1, 5)
- ○ (3, 5)
- ○ (2, −4)

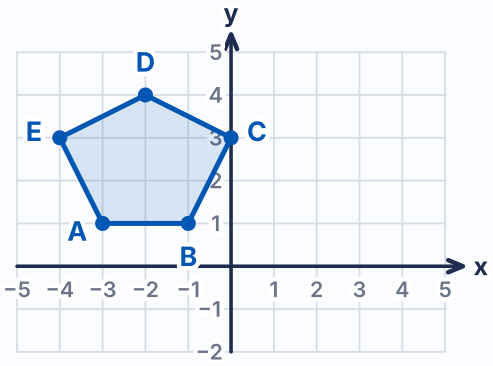



**Rule: (x, y) → (x + 2, y − 4)**

**blue** = pre-image · **gold dashed** = image

**P2** **Pentagon ABCDE has vertex A(−3, 1). The whole pentagon is translated by (x + 5, y − 2). Write the coordinates of A′.**

A′ = ( ____ , ____ )

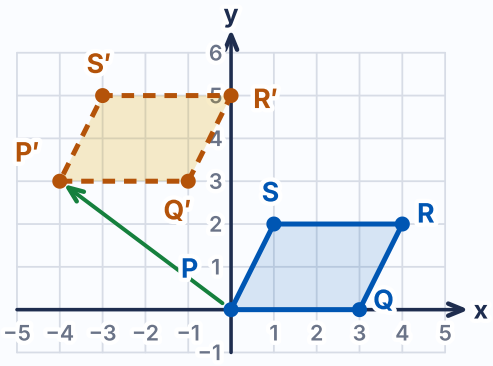



**Rule: (x, y) → (x + 5, y − 2)**

**blue** = pre-image · **gold dashed** = image

**P3** **Parallelogram PQRS was translated to P′Q′R′S′ (shown). P(0, 0) maps to P′(−4, 3). Which rule was used?**

- ○ (x − 4, y + 3)
- ○ (x + 4, y − 3)
- ○ (x − 3, y + 4)
- ○ (x + 4, y + 3)




**blue** = pre-image · **gold dashed** = image

## 3 Reflections (Flips)

📌 **Reflection:** Every point flips across a **line of reflection** (the mirror line). Each point lands the same distance from the mirror, on the opposite side.

📌 **NC Standard 8.G.3:** 8th grade focuses on reflections across the **x-axis** and **y-axis**.

💡 **One more mirror line:** To reflect over the line **y = x**, swap the numbers: **(x, y) → (y, x)**. Example: (1, 4) → (4, 1).

### The Rules

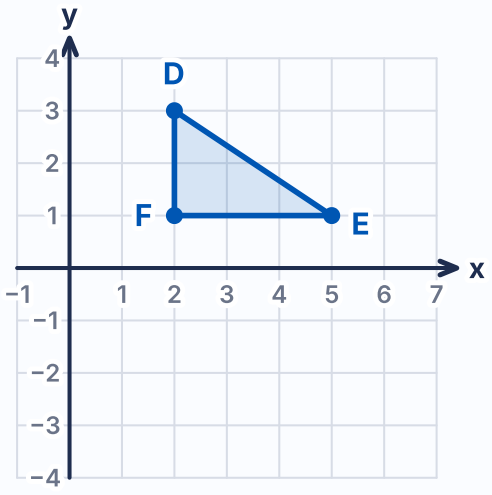
| Line of Reflection | Rule | Example: (3, 4) → | What Changes |
|---|---|---|---|
| **x-axis** | `(x, y) → (x, −y)` | (3, −4) | y flips sign |
| **y-axis** | `(x, y) → (−x, y)` | (−3, 4) | x flips sign |

📌 **Memory trick:** The coordinate that *matches the axis name* is the one that **stays the same**. Reflect over the x-axis → x stays. Reflect over the y-axis → y stays.

## Worked Example

### Example 2: Reflect triangle DEF with D(2, 3), E(5, 1), F(2, 1) over the x-axis.

*Pre-image △DEF (mirror = x-axis).*




**Rule: (x, y) → (x, −y)**

**blue** = pre-image · **gold dashed** = image

**Step 1 — Identify the rule:** _____

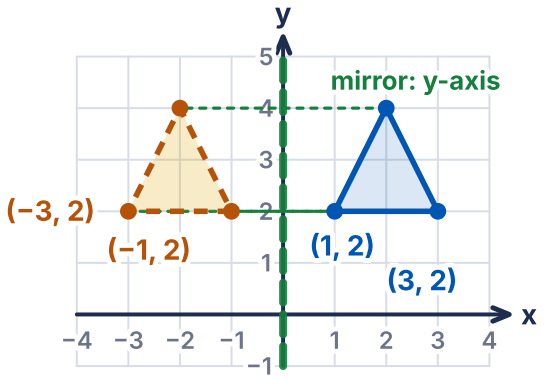
**Step 2 — Apply to D(2, 3):** _____

**Step 3 — Apply to E(5, 1):** _____

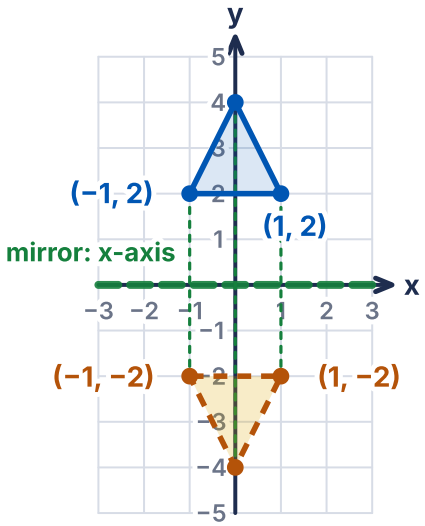
**Step 4 — Apply to F(2, 1):** _____

✓ **Image vertices:** _____

## Visual — Two Mirrors Side by Side




Reflect over the y-axis: x flips sign



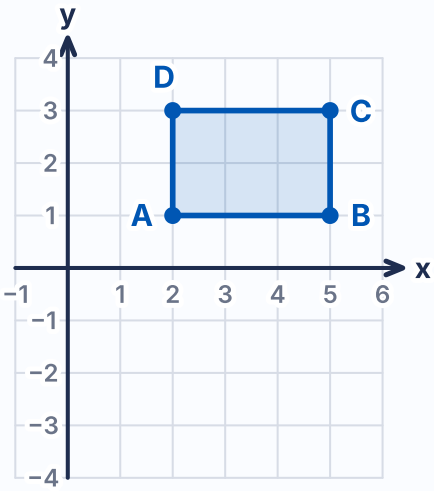

Reflect over the x-axis: y flips sign

**blue** = pre-image · **gold dashed** = image

## Practice

**P1** Rectangle ABCD has vertex C(5, 3). The whole rectangle is reflected over the x-axis. What are the coordinates of C′?

- ○ (5, −3)
- ○ (−5, 3)
- ○ (−5, −3)
- ○ (3, 5)

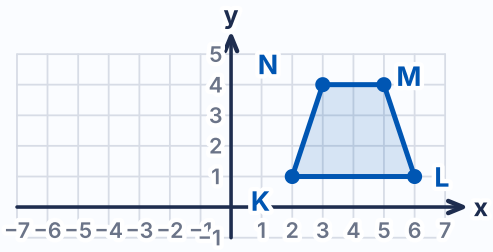



**Rule: (x, y) → (x, −y)**

**blue** = pre-image · **gold dashed** = image

**P2** **Trapezoid KLMN has vertex M(5, 4). The whole trapezoid is reflected over the y-axis. Write the coordinates of M′.**

M′ = ( ____ , ____ )

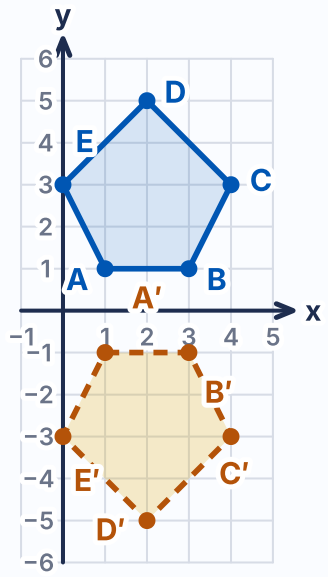



**Rule: (x, y) → (−x, y)**

**blue** = pre-image · **gold dashed** = image

**P3** **Pentagon ABCDE was reflected to A′B′C′D′E′ (shown). Which line is the mirror (line of reflection)?**

- ○ the x-axis
- ○ the y-axis
- ○ the line y = x
- ○ the line x = 2




**blue** = pre-image · **gold dashed** = image

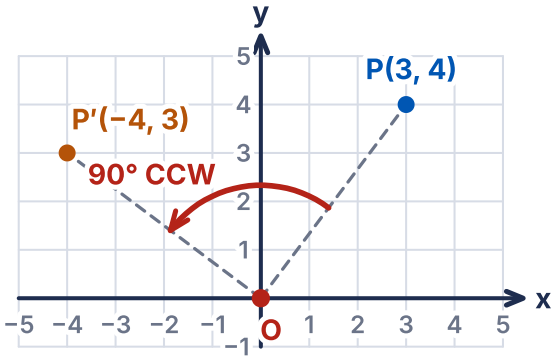
## 4 Rotations (Turns)

📌 **Rotation:** Every point turns a fixed angle around a **center of rotation**. In this unit, the center is always the **origin (0, 0)**.

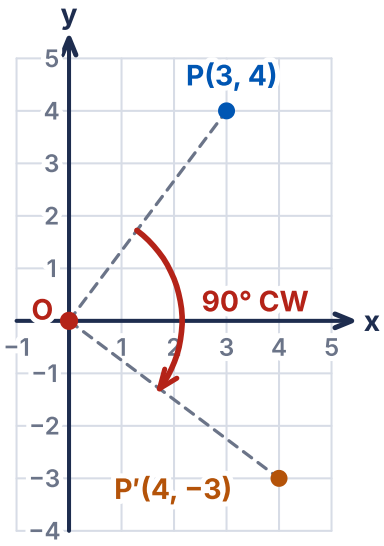
📌 **NC Standard 8.G.3:** 8th grade focuses on rotations of **90°, 180°, and 270°** about the origin.

↻ **Direction matters:** A rotation can be **counterclockwise (CCW)** or **clockwise (CW)**. Always check which way the problem asks. **270° one way = 90° the other way.** 180° is the same either way.

## The Rules (About the Origin)




90° counterclockwise




90° clockwise

Same point, same 90° turn about O, opposite directions. The red arrow shows the direction.

Learn these three rules.

| Rotation | Rule | Example: (3, 4) |
|---|---|---|
| **90° CCW** | $(x, y) \to (-y, x)$ | (−4, 3) |
| **180°** | $(x, y) \to (-x, -y)$ | (−3, −4) |
| **90° CW** | $(x, y) \to (y, -x)$ | (4, −3) |

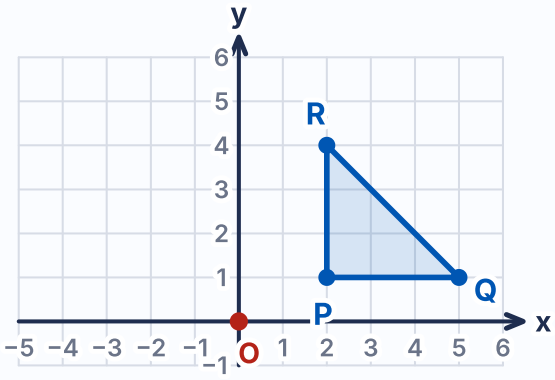
📌 **270°?** 270° CCW is the same as 90° CW. 270° CW is the same as 90° CCW.

⚠ **Watch out:** 180° is the only rotation where the coordinates are NOT swapped. For 90° and 270°, you must swap x and y first, THEN apply the sign rule.

## Worked Example

**Example 3: Rotate triangle PQR with P(2, 1), Q(5, 1), R(2, 4) 90° counterclockwise about the origin.**

*Pre-image △PQR and the 90° counterclockwise turn about the origin.*




**Rule: (x, y) → (−y, x)**

**blue** = pre-image · **gold dashed** = image

**Step 1 — Identify the rule:** _____

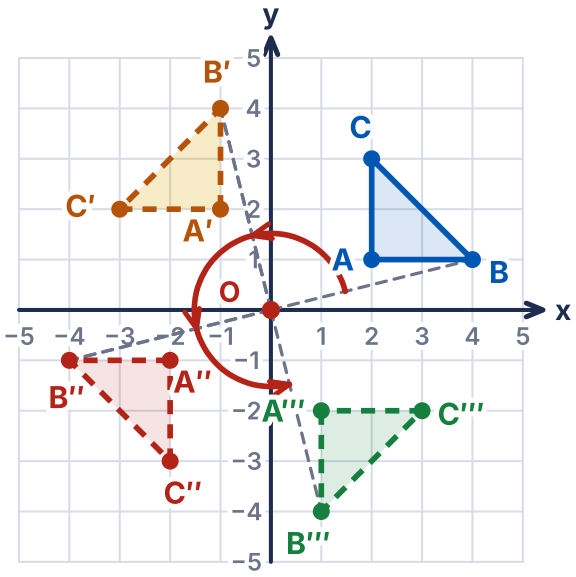
**Step 2 — Apply to P(2, 1):** _____

**Step 3 — Apply to Q(5, 1):** _____

**Step 4 — Apply to R(2, 4):** _____

✓ **Image vertices:** _____

## Visual — One Triangle, Three Rotations




↻ each red arrow = one 90° CCW turn

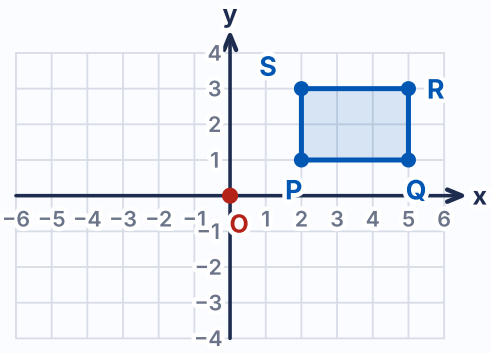
**△ABC → 90° CCW → 180° → 270° CCW (= 90° CW)**. Each red arrow is one more 90° turn about O.

*Same triangle △ABC, three different rotations. Notice the triangle stays the same size and shape — only where it sits and which way it faces change.*

## Practice

**P1** **Rectangle PQRS has vertex R(5, 3). The whole rectangle is rotated 180° about the origin. What are the coordinates of R′?**

- ○ (−5, −3)
- ○ (5, −3)
- ○ (−3, −5)
- ○ (3, 5)

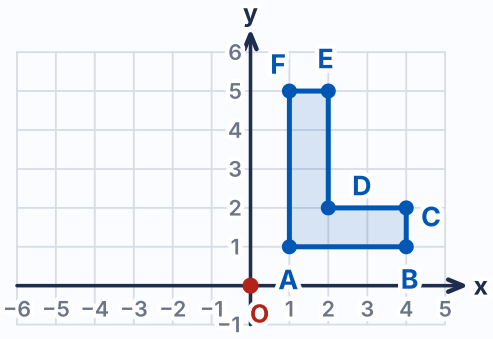



**Rule: (x, y) → (−x, −y)**

**blue** = pre-image · **gold dashed** = image

**P2** **L-shape ABCDEF has vertex E(2, 5). The whole shape is rotated 90° counterclockwise (CCW) about the origin. Write the coordinates of E′.**

E′ = ( ____ , ____ )

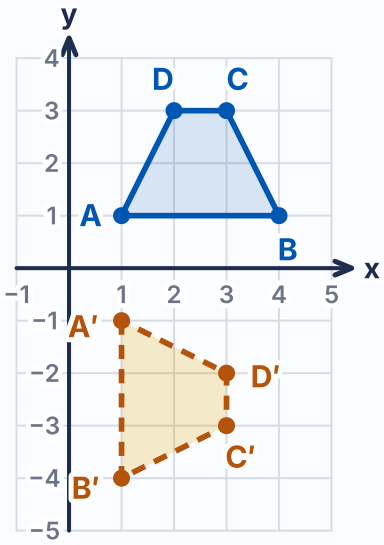



**Rule: (x, y) → (−y, x)**

**blue** = pre-image · **gold dashed** = image

**P3** **Trapezoid ABCD was rotated to A′B′C′D′ (shown). All rotations are about the origin. Which rotation was used?**

- ○ 90° clockwise (CW)
- ○ 90° counterclockwise (CCW)
- ○ 180°
- ○ 270° clockwise (CW)




**blue** = pre-image · **gold dashed** = image

## 5 Dilations (Resize)

📌 **Dilation:** A transformation that **resizes** a figure by a **scale factor k** from a center point (usually the origin). The shape stays the same; the size changes.

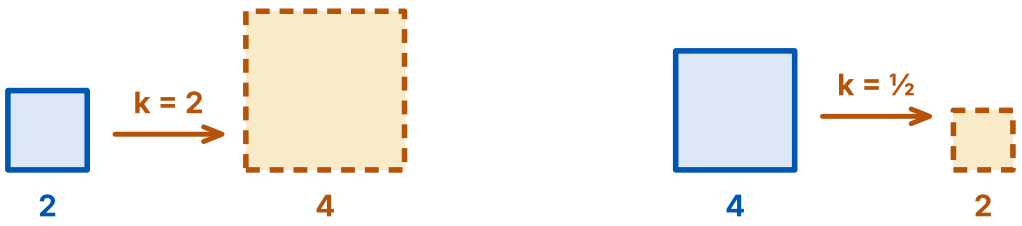
📌 **NC Standard 8.G.4:** Dilations in 8th grade are centered at the origin. Scale factor k can be a whole number, decimal, or fraction.

## The Rule (Centered at the Origin)

$$(x, y) \rightarrow (kx, ky)$$

Multiply *both* coordinates by the scale factor **k**.

## What k Tells You




k > 1 → bigger (enlargement) · 0 < k < 1 → smaller (reduction) · k = 1 → same size

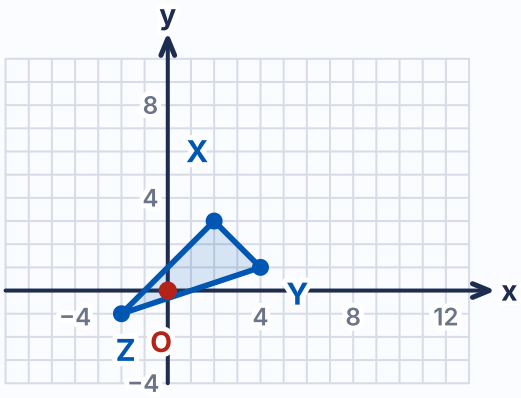
| Scale Factor k | Effect | Image vs. Pre-image |
|---|---|---|
| **k > 1** | Enlargement | Image is **bigger** |
| **k = 1** | No change | Same size |
| **0 < k < 1** | Reduction | Image is **smaller** |

⚠ **Watch out:** Dilation is the ONLY transformation in this unit that is NOT rigid. The pre-image and image are SIMILAR, but NOT congruent (unless k = 1).

## Worked Example

**Example 4: Triangle XYZ has X(2, 3), Y(4, 1), Z(−2, −1). Dilate by scale factor k = 3 (centered at origin).**

*Pre-image △XYZ with rays from the center (0, 0).*




**Rule: (x, y) → (3x, 3y)**

**blue** = pre-image · **gold dashed** = image

**Step 1 — Identify the rule:** _____

**Step 2 — Apply to X(2, 3):** _____

**Step 3 — Apply to Y(4, 1):** _____

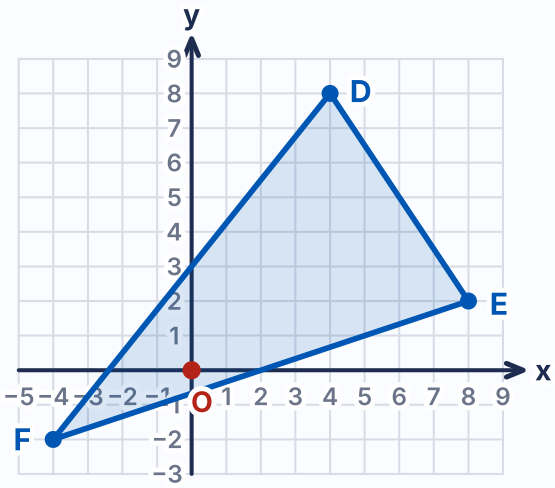
**Step 4 — Apply to Z(−2, −1):** _____

✓ **Image vertices:** _____

## Worked Example — Reduction

**Example 5: Triangle DEF has D(4, 8), E(8, 2), F(−4, −2). Dilate by scale factor k = $\frac{1}{2}$ (centered at origin).**

*Pre-image △DEF with rays from the center (0, 0).*




**Rule: (x, y) → (½x, ½y)**

**blue** = pre-image · **gold dashed** = image

**Step 1 — Rule:** _____

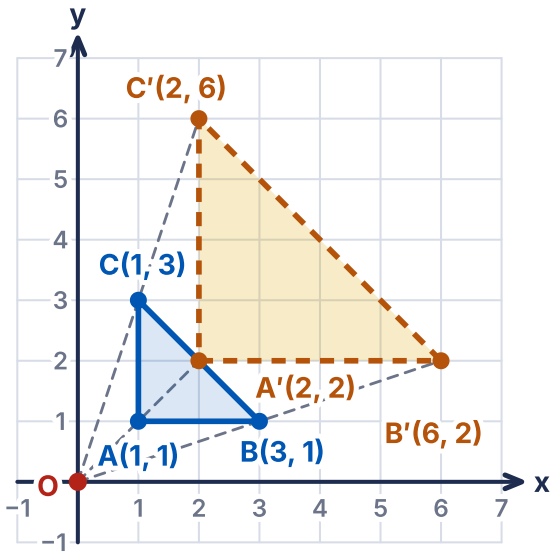
**Step 2 — Apply to D(4, 8):** _____

**Step 3 — Apply to E(8, 2):** _____

**Step 4 — Apply to F(−4, −2):** _____

✓ **Image vertices:** _____

## Visual — Dilation by k = 2




**Rule: (x, y) → (2x, 2y)**

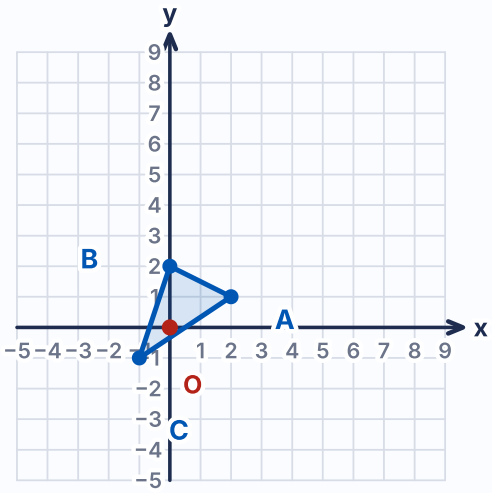
**blue** = pre-image · **gold dashed** = image

Scale factor k = 2 (enlargement): each image point is twice as far from O along the same ray.

## Practice

**P1** Triangle ABC has vertices A(2, 1), B(0, 2), C(−1, −1). The triangle is dilated by scale factor k = 4 (centered at origin). What are the new coordinates of A′?

- ○ (8, 4)
- ○ (6, 5)
- ○ (2, 4)
- ○ (0.5, 0.25)

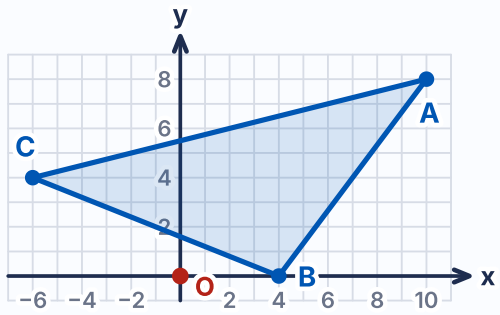



**Rule: (x, y) → (4x, 4y)**

**blue** = pre-image · **gold dashed** = image

**P2** Triangle ABC has vertices A(10, 8), B(4, 0), C(−6, 4). The triangle is dilated by k = $\frac{1}{2}$ (centered at origin). Write the new coordinates of vertex A′:

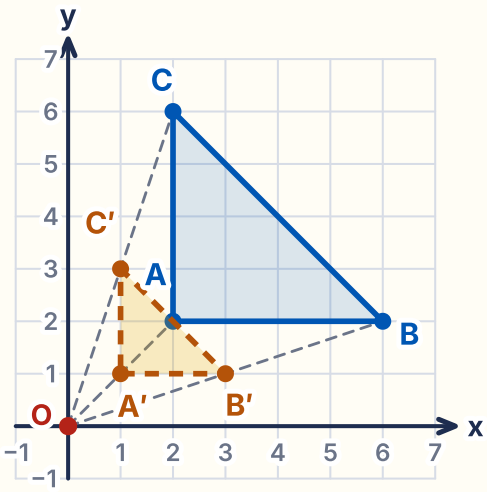
A′ = ( ____ , ____ )




**Rule: (x, y) → (½x, ½y)**

**blue** = pre-image · **gold dashed** = image

**P3** A figure is dilated and the image is *smaller* than the pre-image. The scale factor k must be...




The **image** is smaller. What must k be?

- ○ greater than 1
- ○ equal to 1
- ○ between 0 and 1
- ○ negative

## 6 Congruence & Similarity

Every transformation we've studied tells us how two figures are related. Here's the big-picture connection:

📌 **Congruent figures (≅):** Same size AND same shape. One can be mapped to the other using only **rigid transformations** (translations, reflections, rotations — or any combination).

📌 **Similar figures (~):** Same shape but possibly different size. One can be mapped to the other using **rigid transformations AND a dilation**.

**The Rule**

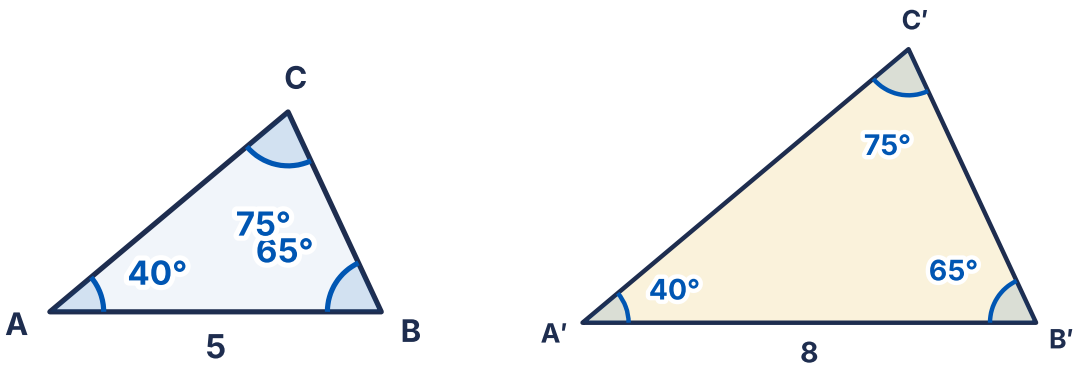
If you can get from Figure A to Figure B using **only translations, reflections, and rotations** → they are **CONGRUENT**.

If you need a **dilation** somewhere in the sequence → they are **SIMILAR** (but not congruent, unless k = 1).

### What Gets Preserved?

| Property | Translations | Reflections | Rotations | Dilations |
|---|---|---|---|---|
| **Side lengths** | ✓ same | ✓ same | ✓ same | × scaled by k |
| **Angle measures** | ✓ same | ✓ same | ✓ same | ✓ same |
| **Shape** | ✓ same | ✓ same | ✓ same | ✓ same |
| **Size** | ✓ same | ✓ same | ✓ same | × changes |

💡 **Notice:** Dilations are the only transformation that changes *side lengths*. Angles stay the same in every transformation — that's why the shape never distorts.

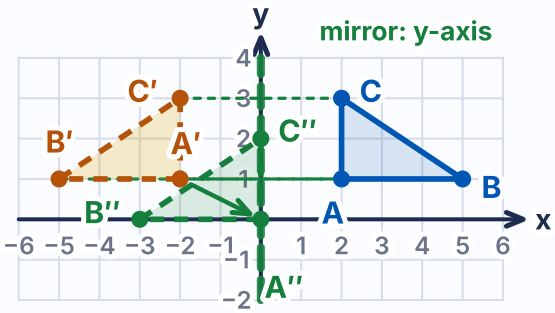



Dilation with k = 1.6: the 40°, 65° and 75° angles stay the same; side 5 becomes 5 × 1.6 = 8.

## Worked Example

**Example 6: Triangle ABC is reflected over the y-axis, then translated by (x + 2, y − 1). Are the pre-image and image congruent, similar, or neither?**

*A sample △ABC through the same two moves.*




**Step 1: (x, y) → (−x, y)**
**Step 2: (x, y) → (x + 2, y − 1)**

**blue** → **gold** (step 1) → **green** (step 2)

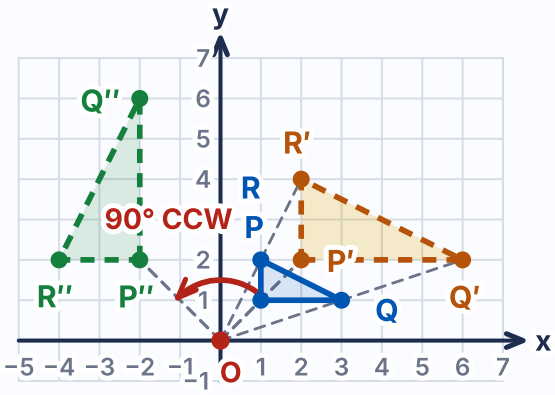
**Step 1 — List the transformations used:** _____

**Step 2 — Any dilations?** _____

✓ **Conclusion:** _____

**Example 7: Triangle PQR is dilated by k = 2, then rotated 90° CCW. Congruent, similar, or neither?**

*A sample △PQR through the same two moves.*




**Step 1: (x, y) → (2x, 2y)**
**Step 2: (x, y) → (−y, x)**

**blue** → **gold** (step 1) → **green** (step 2)

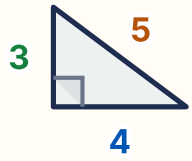
**Step 1 — List the transformations:** _____

**Step 2 — Was a dilation used (with k ≠ 1)?** _____

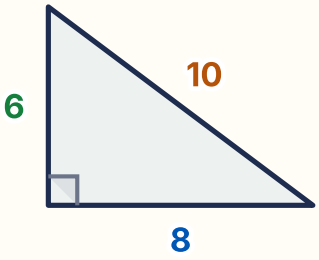
✓ **Conclusion:** _____

## Practice

**P1** Triangle A has side lengths 3, 4, 5. Triangle B has side lengths 6, 8, 10 and the same angles. Triangles A and B are...



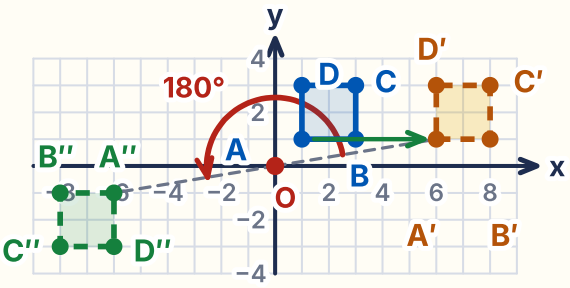

Triangle A




Triangle B

- ○ Congruent
- ○ Similar
- ○ Neither

**P2** A square is translated 5 units right, then rotated 180° about the origin. The pre-image and image are...

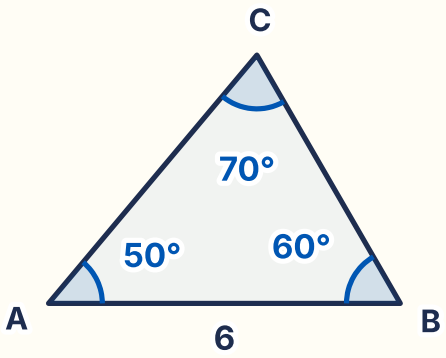



**Step 1: slide 5 right**
**Step 2: turn 180°**

**blue** → **gold** (step 1) → **green** (step 2)

- ○ Congruent
- ○ Similar but not congruent
- ○ Neither

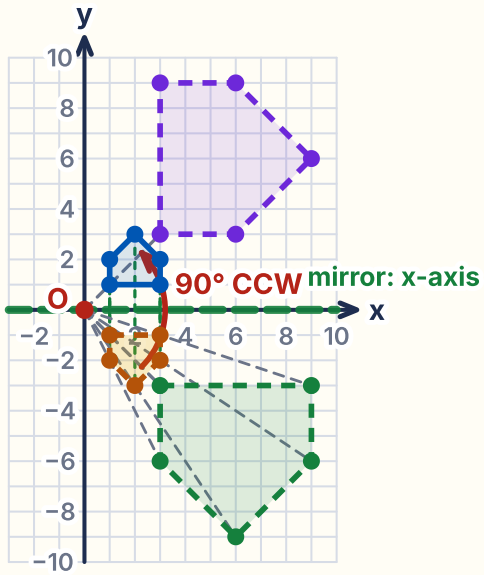
**P3** **Which transformation preserves angle measure but NOT side length?**




Which move keeps all three angles but changes the side 6?

- ○ Translation
- ○ Reflection
- ○ Rotation
- ○ Dilation

⭐ Challenge **A pentagon undergoes a sequence: reflection over the x-axis, dilation by k = 3, then rotation 90° CCW. The pre-image and image are...**




**Step 1: flip over x-axis**
**Step 2: dilate k = 3**
**Step 3: turn 90° CCW**

**blue** → **gold** (step 1) → **green** (step 2) → **purple** (step 3)

- ○ Congruent
- ○ Similar (not congruent)
- ○ Neither

## Unit 8 Wrap-Up

You've now seen all four transformation types and how they connect to congruence and similarity. Quick recap:

- **Translation:** (x, y) → (x + h, y + k) — slide. Rigid.
- **Reflection over x-axis:** (x, y) → (x, −y) — y flips. Rigid.
- **Reflection over y-axis:** (x, y) → (−x, y) — x flips. Rigid.
- **Reflection over y = x:** (x, y) → (y, x) — swap. Rigid.
- **Rotation 90° CCW:** (x, y) → (−y, x). Rigid.

- **Rotation 180°:** (x, y) → (−x, −y). Rigid.
- **Rotation 90° CW (same as 270° CCW):** (x, y) → (y, −x). Rigid.
- **Dilation (k from origin):** (x, y) → (kx, ky). Non-rigid.

💡 **Next step:** Head to the Unit 8 page to try the practice problems and the Transformation Quest game.